

# Nithin Govindarajan

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Location: Leuven, Belgium

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## Areas of expertise

Numerical linear algebra, Numerical analysis, systems theory, dynamical systems and control, signal processing, Computational algebraic geometry, PDE solvers.

## Education

- Oct. 2014 - Dec. 2018 PHD in Mechanical Engineering, University of California in Santa Barbara  
*Dissertation title:* “Periodic approximations and spectral analysis of the Koopman operator: theory and applications”.  
*Advisors:* I. Mezić, S. Chandrasekaran.
- Sep. 2009 - Oct. 2012 MSC in Aerospace Engineering (with distinction), Technische Universiteit Delft  
*Master thesis:* “An Optimal Control Approach for Estimating Aircraft Command Margins”.  
*Advisors:* Q.P. Chu, C.C. de Visser.
- Sep. 2006 - Jul. 2009 BSC in Aerospace Engineering (with distinction), Technische Universiteit Delft

## Work experience

- Nov. 2023 - present Founder of VGN Computational consultancy services
- Nov. 2018 - present Co-founder and technical brain of software startup [www.matisse.ai](http://www.matisse.ai) in dentistry
- Aug. 2019 - Feb. 2026 Postdoctoral researcher, KU Leuven, Belgium
- Feb. 2019 - July 2019 Lecturer Mathematics, University of Amsterdam (UvA), The Netherlands
- Nov. 2012 - May 2013 Junior R&D engineer, National Aerospace Laboratory (NLR), Amsterdam, The Netherlands
- Sep. 2011 - May 2012 Intern, Mission Critical Technologies Inc. (on site at NASA Ames), Moffet Field, CA

## Fellowships, honors & awards

- Apr. 2016 CCDC fellowship, Center for Control, Dynamics and Computation, Santa Barbara, CA
- Apr. 2014 Department Merit Fellowship (UCSB Mech. Eng), Santa Barbara, CA
- May 2011 Huygens Scholarship Programme, Nuffic, The Hague, The Netherlands

## Publications

### Journal Publications

- 2025 Govindarajan, N., Chandrasekaran, S., Dewilde Lathauwer, L. (2025). Tree Quasi-separable Matrices: A Simultaneous Generalization of Sequentially and Hierarchically Semiseparable Representations. *SIAM Journal on Matrix Analysis and Applications*, 46(2), 1562-1586.
- 2024 Govindarajan, N., Widdershoven, R., Chandrasekaran, S., & De Lathauwer, L. (2023). A

fast algorithm for computing Macaulay nullspaces of bivariate polynomial systems. *SIAM Journal on Matrix Analysis and Applications*, 45(1), 368-396.

Govindarajan, N., Vervliet, N., & De Lathauwer, L. (2022). Regression and classification with spline-based separable expansions. *Frontiers in big Data*, 5, 688496.

Govindarajan, N., Epperly, E. N., & Lathauwer, L. D. (2022).  $(L_r, L_r, 1)$ -Decompositions, Sparse Component Analysis, and the Blind Separation of Sums of Exponentials. *SIAM Journal on Matrix Analysis and Applications*, 43(2), 912-938.

Epperly, E. N., Govindarajan, N., & Chandrasekaran, S. (2021). Minimal rank completions for overlapping blocks. *Linear Algebra and its Applications*, 627, 185-198.

Govindarajan, N., Mohr, R., Chandrasekaran, S., & Mezic, I. (2021). On the approximation of Koopman spectra of measure-preserving flows. *SIAM Journal on Applied Dynamical Systems*, 20(1), 232-261.

Govindarajan, N., Mohr, R., Chandrasekaran, S., & Mezic, I. (2019). On the approximation of Koopman spectra for measure preserving transformations. *SIAM Journal on Applied Dynamical Systems*, 18(3), 1454-1497.

Govindarajan, N., De Visser, C. C., Van Kampen, E., Krishnakumar, K., Barlow, J., & Stepanyan, V. (2015). Optimal control framework for estimating autopilot safety margins. *Journal of Guidance, Control, and Dynamics*, 38(7), 1197-1207.

Govindarajan, N., de Visser, C. C., & Krishnakumar, K. (2014). A sparse collocation method for solving time-dependent HJB equations using multivariate B-splines. *Automatica*, 50(9), 2234-2244.

### Conference proceedings

Widdershoven, R., Govindarajan, N., De Lathauwer, L. (2023, September). Overdetermined systems of polynomial equations: tensor-based solution and application. *Proceedings of EUSIPCO 2023*, Helsinki, Finland.

Chandrasekaran, S., Govindarajan, N., & Rajagopal, A. (2018, July). Fast Algorithms for Displacement and Low-Rank Structured Matrices. In *Proceedings of the 2018 ACM International Symposium on Symbolic and Algebraic Computation* (pp. 17-22).

Govindarajan, N., Arbabi, H., Van Blargian, L., Matchen, T., & Tegling, E. (2016, December). An operator-theoretic viewpoint to non-smooth dynamical systems: Koopman analysis of a hybrid pendulum. In *2016 IEEE 55th Conference on Decision and Control (CDC)* (pp. 6477-6484). IEEE.

### Preprints & Tech reports

Widdershoven, R., Govindarajan, N., De Lathauwer, L. (2025). Fast Macaulay null space through the intersection of shifted null spaces. Technical Report 25-56, ESAT-STADIUS, KU Leuven.

Chandrasekaran, S., Epperly, E. N., Govindarajan, N. (2019). Graph-induced rank structures and their representations. *arXiv preprint arXiv:1911.05858*.

### Selected talks

“Macaulay matrices, low displacement rank, and the efficient computation of null spaces”, Back to the roots seminar, Leuven.

“A tensor-based approach to solving systems of multivariate polynomials”, CAM23, Selva di Fasano.

“Efficient Computation of Macaulay Matrix Null Spaces Through Exploiting Shift-Invariant Structures”, SIAM AG23, Eindhoven.

“ $(L_r, L_r, 1)$ -decompositions, Sparse Component Analysis, and the Blind Separation of Sums

|      |                                                                                                                                                                                                                                        |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2018 | of Exponentials”, SeLMA meeting, Leuven. “Spline-based separable expansions for approximation, regression and classification”, IPAM Workshop I: Tensor Methods and their Applications in the Physical and Data Sciences, UCLA (online) |
| 2017 | “A toolbox for computing spectral properties of dynamical systems”, SIAM DS17, Snowbird.                                                                                                                                               |
| 2016 | “An operator-theoretic viewpoint to non-smooth dynamical systems: Koopman analysis of a hybrid pendulum”, IEEE CDC 16, Las Vegas.                                                                                                      |

## Teaching

### Lead instructor & course organizer

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Semester 1 2019/2020 | Numerical mathematics, Amsterdam University College |
|----------------------|-----------------------------------------------------|

### Co-instructor

|                      |                                                          |
|----------------------|----------------------------------------------------------|
| March 2022           | Fast algorithms for dense structured matrices, KU Leuven |
| Semester 2 2021/2022 | Numerieke modellering & benadering, KU Leuven            |
| Semester 2 2019/2020 | Numerieke modellering & benadering, KU Leuven            |

### Teaching assistant

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|----------------------|-------------------------------------------------------------------|
| Semester 1 2021/2022 | Numerieke wiskunde, KU Leuven                                     |
| Semester 1 2020/2021 | Numerieke wiskunde, KU Leuven                                     |
| Semester 1 2019/2020 | Numerieke wiskunde, KU Leuven                                     |
| Spring 2018          | Control theory, UCSB                                              |
| Winter 2017          | Electrical circuits Lab, UCSB                                     |
| Fall 2017            | Intro to programming in Matlab, UCSB                              |
| Summer 2017          | Dynamics, UCSB                                                    |
| Summer 2017          | Physics Lab: intro to classical mechanics for non-engineers, UCSB |
| Spring 2017          | Dynamics, UCSB                                                    |
| Spring 2015          | Dynamics, UCSB                                                    |
| Winter 2015          | Vibrations, UCSB                                                  |
| Fall 2014            | Statics, UCSB                                                     |

## Extra-curricular

### Software skills

Matlab, Python, Julia, Mathematica (basic), C++ (basic), Latex, Git.

### Languages

English, Dutch, Tamil (basic).

Last updated: February 16, 2026